

PneuMat: Pneumatic Interaction System for Infant Sleep Safety Using Shape-Changing Interfaces

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ABSTRACT

Sleep plays an integral role in human health and is vitally important for neurological development in infants. In this study, we propose the PneuMat, an interactive shape-changing system integrating sensors and pneumatic drives, to help ensure sleep safety through novel human-computer interaction. This system comprises sensor units, control units and inflatable units. The sensor units utilize information exchange between infants and the system, and detect the infant's sleeping posture, sending raw data to control units. For better sleep experience, the inflatable units are divided into nine areas. The inflatable units are multi-mode, can be independently inflated in different areas, and can be inflated in different areas together. We aim to ensure sleep safety by ensuring that infants stay in a safe sleeping position while in bed, by autonomously actuating the PneuMat's shape-changing capability. In this article, we describe the division of PneuMat, the design of the control unit, integration of the sensors and our preliminary experiments to evaluate the feasibility of our interaction system. Finally, based on the results, we will discuss future work involving the PneuMat.

CCS CONCEPTS

• **Human-centered computing**; • **Human computer interaction (HCI)**;

KEYWORDS

Interaction system, Infant, Shape-changing interfaces

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1 INTRODUCTION

The Early Child Care Research Network reports that there are about 3,500 sleep-related deaths among US infants each year. The sleeping position of infants is very important as it directly affects growth and safety. The wrong sleeping position can cause problems such as deformational plagiocephaly (DP), a condition in which deformation of the infant's skull occurs in the first months of life, caused by different reasons, including a preference for a particular sleeping position [5]. Worst of all, the wrong sleeping position can lead to infant death. One study found that two out of 15 reported infant deaths involved infants having been placed in a prone position, and 13 (87%) had rolled into a prone position from either the supine or side position [3]. Studies have found that a side sleeping position is unstable and increases the chance that infants will roll onto their stomachs – the sleep position associated with the highest risk of sudden infant death syndrome (SIDS) [1]. Thus, it is important to keep the infant in a safe position during sleep.

In human-computer interaction (HCI), researchers have explored a variety of solutions for information exchange and human-computer interaction with infants during their sleep time, such as using deep learning techniques to automatically assess photographs of the infant's sleeping environment [4], and detecting the components of exhaled breath during sleep using a novel, tiny gas sensor [14], etc. Most of these project alert caregivers, rather than autonomously taking action themselves when infants are in an unsafe environment. Our interactive system, the PneuMat (Figure 1), has integrated sensors and inflatable units, and can detect and autonomously adjust the infant's posture before and during sleep so that parents and caregivers do not need to be on standby for 24 hours.



Figure 1: Demonstration of the PneuMat

2 RELATED WORKS

2.1 Infant-related works in HCI

Infants are becoming a large user group for new technologies around the world. We believe that it is essential for the HCI community to continue taking the lead in providing children with the best interactive system. In the HCI community, there has been a recent trend focusing on novel computing-based research for infants. Some of these studies have reduced infants' negative experiences and promoted health. For example, Calmer [13] (proposed by Hauser et al.) reduces premature babies' pain by as much as 50%. InfantTracks [2] (proposed by Ayelet Ben-Sasson et al.) and studies by Hayes et al. [6] provide support for the transition from hospital to home for premature infants. The Smart Spoon [8] (proposed by Geeyoung Noh et al.) monitors how healthy infants' diets are. Some of these technologies focus on the emotional interaction between infants and their parents, e.g., the Mimo Pillow [15], designed by Versteegh and Chen to comfort newborn infants. The Family Arizing System [12] (proposed by Croes et al.) enables parents to send "hugs" to their infants remotely. In this study, we chose to focus on sleep safety of infants under 6 months, and we are proposing a pneumatic interaction system.

2.2 Shape-changing with sensing techniques

In the field of human-computer interaction, there are many studies on technologies to build shape-changing interfaces, and the feasibility of shape-changing infant mats has been proven: PneuUI [18] introduced pneumatically-actuated shape-changing interfaces with integrated sensing; the Aeromorph [10] is a project that focuses on fabrication-transforming inflatables with pneumatic hinges; and Printflatables [9] is another design and fabrication system to make large, functional and dynamic pneumatic artifacts.

A range of sensing techniques for shape change have been explored in HCI. Kim et al. [11] used a touch sensor to track the touch area of the hand, and used the input signal and pressure sensor's sensitivity to expand and contract the balloon; Gohlke et al. [7] arranged pressure sensors on the surface in a lattice for multi-touch and multi-pressure input; Guanyun Wang et al. [16] presented a printed paper actuator as a low cost, reversible and electrical actuation and sensing method; Lingning Yao et al.

[17] introduced the living *Bacillus Subtilis natto* cell as a humidity sensitive nanoactuator; Xinlei Zhang et al. [19] used the electromyography (EMG) sensor to detect the signal of the forearm to pneumatically control the inflatable units placed in the palm of the hand. However, these actuations of shape change rely on conscious and active control, which is obviously unsuitable for infants. When an infant is alone, the autonomous adjustment system with sensing techniques becomes essential because the infant will not yet have mastered the ability to express and control other tools.

3 SYSTEM ARCHITECTURE

The PneuMat integrates sensors and pneumatic drivers, as Figure 2 shown, keeping infants in a safe position while in bed, with no need for parents and caregivers to be around. The pressure sensor, temperature sensor and ultrasonic sensor read the pressure, temperature and distance between the infant and the cushion respectively. And liquid sensor is used to detect whether infant sudden onset acute and severe choking milk. All sensors provide sensory input to the system so that the PneuMat can autonomously adjust a dangerous sleeping posture by shape-changing. Figure 3 shows a block diagram of the complete system architecture and signal flow.

Pre-inflation: After being fed, infants are usually placed on cushions to rest. Before the infant is placed on the mat, the PneuMat runs an inflation procedure in advance (area "A" (Figure 4) is in state III) so that the infant is placed on a inclined plane, and this position will be maintained for 15 minutes. This process is to prevent the infant from choking on milk.

Prevention of a prone position: Temperature sensors and pressure sensors are placed on both sides of the PneuMat. A temperature sensor and a pressure sensor are placed on the inflatable unit that the infant will come into contact with when lying on his side. If the infant's posture is not correct, the sensor will send a signal to the control unit. The inflatable units on the opposite side will inflate tidally to adjust the infant's sleeping position.

The inflatable units in different areas can cooperate in real time. In other situations, when the sensor detects that the infant is maintaining an inappropriate sleep posture, the inflatable units will cooperate with the other inflatable units to adjust the infant's posture slowly.

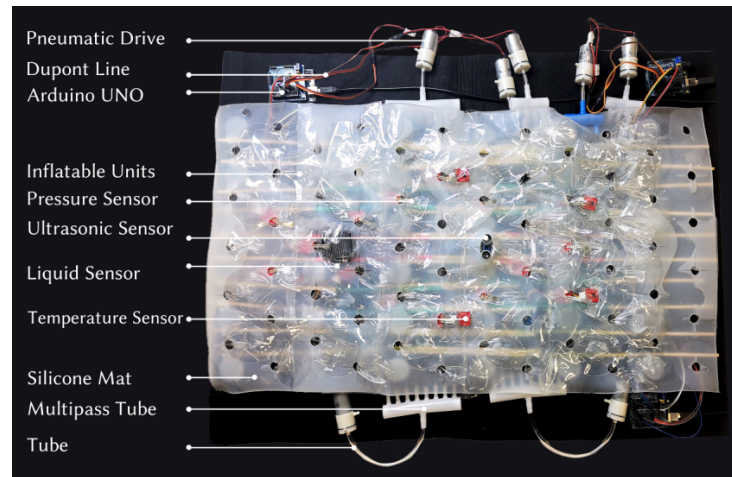


Figure 2: Components of the PneuMat system. The PneuMat is composed of four main parts: (1) nine groups of inflatable units connected by tubes so that the infant can be placed (depending on the size of the object) on the inflatable units; (2) a control unit; (3) a group of integrated analog digital signal sensor units; (4) the computer works remotely for real-time data acquisition and processing. The control units can be attached to the side of the PneuMat via tubes.

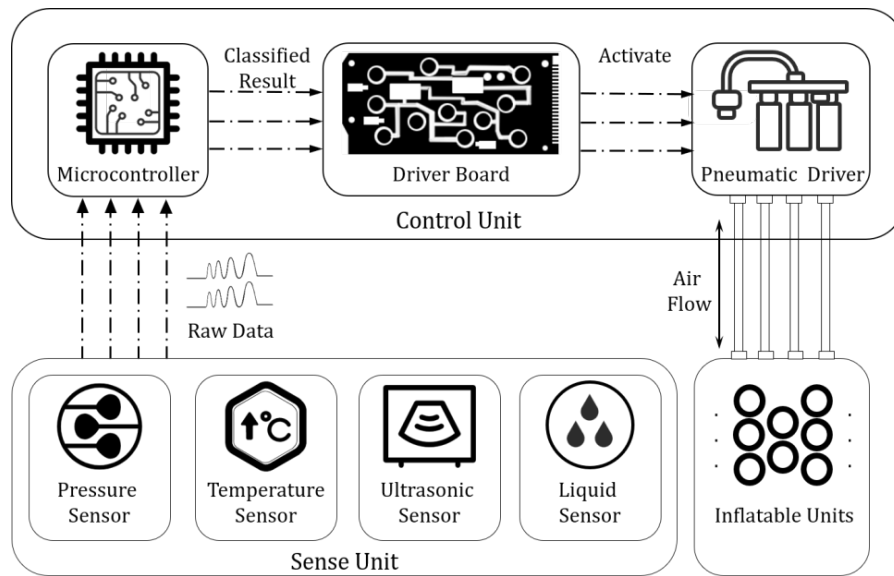


Figure 3: System structure. Each sensor detects movement by the infant and sends out a signal, which is passed to the Arduino to control the pneumatics driving the inflatable units.

3.1 Division of inflatable units

The PneuMat is divided into nine specific areas (Figure 4). Division of the PneuMat based on the infant's body structure to achieve the purpose of more precise adjustment of the infant's sleeping posture and a better interactive experience. Areas A and B are located above the infant's neck, and their function is to maintain the infant's head posture. Through different degrees of inflation, the head can be lifted vertically, or the head can be adjusted horizontally. Area E corresponds to the infant's back; areas C and D correspond to the infant's upper limbs; H and I correspond to the infant's lower limbs;

area F corresponds to the infant's buttocks; and area G corresponds to the thigh area below the infant's buttocks.

Each area is composed of several identical small inflatable units, and each small inflatable unit is inflated to inflate the entire area, which can make the effect of shape-changing more uniform. The inflatable units consist of rubber balloons with a maximum diameter of 70mm. The rubber balloon was chosen because its maximum inflated size matches the expected height of the infant to be lifted. The overall size of the PneuMat is 340 mm x 755 mm. The time

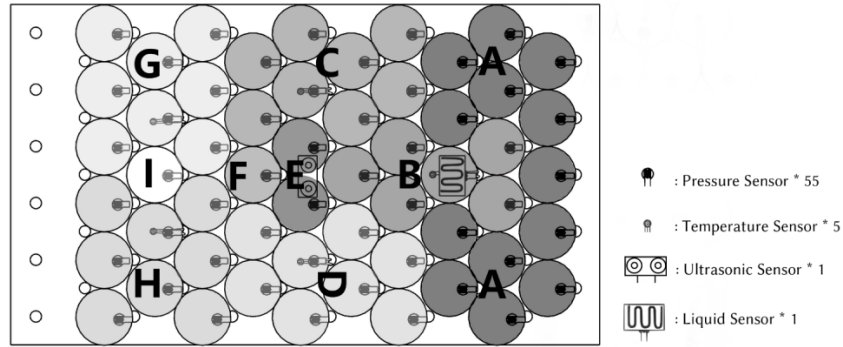


Figure 4: Division of the PneuMat

Table 1: The time taken for each area to reach different levels of inflation (s)

Area	State I	State II	State III
A	0	10.67	21.63
B	0	5.43	8.22
C/D	0	6.85	12.44
E/ F/ I	0	2.28	4.58
G/H	0	6.73	13.83

State I is not inflatable; state II: the height of the expansion unit is 30 mm; state III: the height of the expansion unit is 75 mm.

taken for each area to reach different levels of inflation is shown in Table 1

3.2 Sensor units

We used digital signals and analog signals from integrated sensors for sleeping position sensing. This involved using a pressure sensor and temperature sensor to collect a nine-channel analog pressure signal with a frequency of 30 Hz, which was sent to the Arduino UNO R3 through a sensor shield V5.0 board. In order to collect data and classify signals, we ran the Arduino IDE on a PC and developed a system programming program to perform the decision-making process. To classify the signal from the pressure sensor, temperature sensor, ultrasonic sensor and liquid sensor, we used an integrated system of four sensor shield V5.0 boards, which contained 55 signal sets. We classified the collected dataset, which contained 5000 records of nine channel digital signals and analog signals data, and every posture-adjustment movement was classified. When preprocessing the data, we first converted the data value of each analog signal into its absolute value, and then processed the corresponding value of the inflatable unit. The same program was used to process the original analog signal data in real time, and then the corresponding prototype construction. Classification of all the process data was carried out in real time.

The sensor units utilize information exchange between infants and the system. Five temperature sensors distributed in five preset locations determine if the infant is in the correct sleeping position

by detecting the temperature signal. The pressure sensor is easily affected by items such as blankets and dolls. The locations and number of each sensor are shown in the Figure 4. The locations of the sensors are based on ergonomics. If the infant is in a dangerous and inappropriate sleeping position, the sensor receives the signal, and then the corresponding pneumatic driver is activated until sensor signal returns to the initial state. Figure 5 illustrates the relationship between the analog signal of the sensor unit and the state of the pneumatic driver.

3.3 Control units

We used an Arduino Uno as the controller for our system. It has power management and built-in voltage regulation, and can be used with a variety of processor chips. We built our own pneumatic circuit, consisting of a miniature pump, a PWM modulator and a solenoid valve converted into a valve array of nine two-way valves. Each valve was 36 mm wide, with a discharge speed < 5S (discharge time from 300 mm/Hg to 15 mm/Hg in a 500-cc tank).

We used the KE2055-1-1 pressure sensor to detect pressure from the infant. The pressure sensor was made of nano-pressure sensitive materials in order to sense the external pressure signal, convert the pressure signal into an electrical signal of corresponding (varying) strength and send this to the controller. For data collection and signal classification, we used a pressure sensor and a temperature sensor, both of which were fixed to the inflatable units. Tidal inflating was based on the inflation and deflation of three or more inflatable units (Figure 6) to convert gravitational potential energy into kinetic energy.

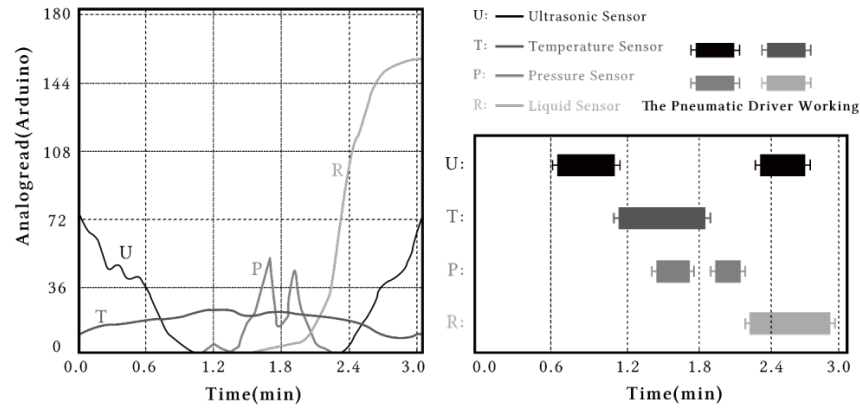


Figure 5: Fluctuations of the analog signal of sensors and status of the pneumatic drivers

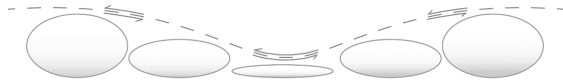


Figure 6: Inflating tidally



Figure 8: Experimental scenarios



Figure 7: The materials used for the casting and casting process: 160 g of organic silicon liquid was poured into module II, and module I was then pressed onto it; the module was placed in the oven to accelerate curing; taking out the silicone mat.

3.4 Fabrication process

A molding and casting method was utilized for fabrication of the PneuMat. Molds were 3-D printed from Unigraphics NX11.0 files (at a high resolution) to create silicone parts with maximally smooth surfaces. SJ3212 silicone part A and part B were used together to form a material with the required module stiffness and elasticity. Components I and II were used to create a plenum with air flow channels. Silicone was poured into component II and placed in the refrigerator for 30min to eliminate air bubbles; it was then put in 122°F oven for 2h to speed up curing. This was repeated six times to mold all the required silicone modules. The products were then sealed together using SJ3212 silicone. Finally, the inflatable units, sensors and pneumatic drives were assembled into the silicone mat. The casting processes are shown in Figure 7

4 PRELIMINARY EVALUATION

4.1 Experiment

To investigate the PneuMat's ability to adjust an infant's sleeping position, we conducted a preliminary test with a baby doll (600

mm in length and 624 g in weight) to adjust sleeping positions in different scenarios. A list of sleeping scenarios and some examples of the device configurations is listed in Figure 8

Before the test, we first confirmed that the interaction between the subject and the shape-changing interfaces performed as expected. During the test, the subject interacted with the system to adjust their sleeping position by firstly lying flat on the PneuMat and then initiating system. All inflatable units were used in the test, placed on the head, neck, trunk and limb regions. Changes in the posture, position and direction in which the subject was lying were allowed if subject could easily be adjusted in this way. Authors were not allowed to help the subject adjust sleeping positions during the tests since, in a real-life situation, intended end-users of the system may not have parents and caregivers present at all times. If the subject's incorrect sleeping position is adjusted and the safe sleeping position is maintained, then the interactive system was considered to be capable of adjusting infant sleeping positions. The experiment was performed for seven categories of sleeping positions: choking on milk, serious choking on milk, a prone position, twisted limbs, tilted head, lying long, and Out off the PneuMat area.

4.2 Results

The results of the experiment, as well as the areas of the inflatable units used to adjust the infant's sleeping position for each scenario, are summarized in Table 2. Ten tests were conducted for each scenario independently. Overall, in most situations, the subject was able to correctly adjust an incorrect sleeping position using the PneuMat. The most effective scenario was "choking on milk". A less effective scenario was "Out off the PneuMat" (requiring the infant to be pushed back into the center of the PneuMat). The inflatable units had a limited contact area with the subject in this particular

Table 2: Preliminary PneuMat-adjustment test results

Scenario	Mapping	Inflatable units	Average time	Success Ratio
Choking on milk	Neck lifted 15 degrees	[I(C-I), II(B), III(A)]	15 min	10/10
Seriously choking	Neck lifted 30 degrees	[I(C-I), III(A, B)]	5 min	8/10
In a prone position	Inflating to block	[I(B, D-I), II(A, C)] / [I(B, C, E-I), II(A, D)]	5 sec	9/10
Twisted limbs	Inflating tidally	[I(A, D, F-I), II(C), III-I(G)] / [I(A-C, E-I), II(D), III-I(H)]	5 sec	7/10
Tilted head	Inflating alternately	[I(C-I), II(B), III(A)]	10 sec	9/10
Lying long	Inflating alternately	[I(E, I), II(B-D, F-H), III(A)]	10 sec	10/10
Out off the PneuMat area	Inflating to block	[I(B, E, F, I), II(C, D, G, H), III(A)]	5 sec	6/10

Average time is PneuMat's working time for each cycle of changing the infant's position to safe position.

sleeping position, so the average success rate in these scenarios was relatively low.

5 LIMITATIONS AND FUTURE WORK

Currently, to use the PneuMat, infants would be in a state of pre-sleep or sleep generally, but the pneumatic drivers of the interactive system will produce some decibels of noise, which may disturb sleep. To address this problem, we plan to have the pneumatic drivers fixed in a box made of sound-absorbing material. The pneumatic drivers of the system could be triggered by an embedded flexible pressure sensor matrix with organic field-effect transistors instead of the pressure sensor currently attached, to create gentler interactive shape-changing system for infants. And as shown from the preliminary test, it is currently difficult to use the PneuMat to adjust positions with a limited contact area. Thus, we plan to increase the number of interactive units of the PneuMat to increase the area of contact between infants and the PneuMat. The weight of the baby doll used in the experiment is much lower than that of a newborn. In addition, the doll was static and remained passive throughout the test. The detection effect of the system on posture changes is uncertain to real newborns. The materials and configurations will have to change a lot to make a working prototype for real conditions. It would be beneficial to collaborate and work together with various stakeholders such as physicians and infants' parents to design, improve and evaluate PneuMat in practice.

6 CONCLUSION

In this paper, we have presented the PneuMat, an interactive shape-changing system integrating sensors and pneumatic drives, which autonomically and independently adjust infants' sleeping position, without the need for parental presence at all times. The PneuMat provides a novel approach to human-computer interaction by using silicone-based inflatable units to achieve shape-changing interface interaction between an infant and the PneuMat. We ran a PneuMat-adjustment test using common sleeping positions to evaluate the PneuMat's ability to adjust sleeping positions. Our results show that the PneuMat can provide reliable, safe position adjustment for most scenarios, except positions with a limited contact area. For future work, we plan to implement a silencing air chamber so that system sound does not disturb infants. A softer, embedded flexible pressure sensor matrix with organic field-effect transistors could be

used to allow the human-computer interaction process to be more comfortable. We should also consider collaborating and working together with various stakeholders such as physicians and infants' parents to design, improve and evaluate PneuMat in practice.

REFERENCES

- [1] American Academy of Pediatrics, Task Force on Infant Positioning and SIDS. Positioning and SIDS: Update. (1996). *Pediatrics*, 98 (6), 1216–1218.
- [2] Ben-Sasson A, Ben-Sasson E, Jacobs K, Rotman Argaman E, and Saig E. 2019. Evaluating Expert Curation in an Infant Milestone Tracking App. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems (CHI '19)*. Association for Computing Machinery, New York, USA, Paper 553, 1–12. DOI:https://doi.org/10.1145/3290605.3300783
- [3] Gerard CM, Harris KA, Thach BT. Physiologic studies on swaddling: an ancient childcare practice, which may promote the supine position for infant sleep. *J. Pediatr.*, 2002, Sep; 141(3): 398–403. doi: 10.1067/mpd.2002.127508. PMID: 12219062.
- [4] Tchuinkou Kwadjo D, Nghonda Tchinda E, Bobda C, Nabaweesi R, Nziengam N, Aitken ME, Whiteside-Mansell L, Barkin S, Mullins SH, and Curran G. 2019. Automatic Assessment of Infant Sleep Safety Using Semantic Segmentation. In *Proceedings of the 13th International Conference on Distributed Smart Cameras (ICDSC 2019)*. Association for Computing Machinery, New York, USA, Article 22, 1–6. DOI:https://doi.org/10.1145/3349801.3349824
- [5] Veloso F *et al*. Synthetic infant head shapes with deformational plagiocephaly: concept and 3D model parameterization, 2019 IEEE 7th International Conference on Serious Games and Applications for Health (SeGAH), Kyoto, Japan, 2019, pp. 1–7. doi: 10.1109/SeGAH.2019.8882444.
- [6] Hayes GR, Patterson DJ, Singh M, Graven D, Rich J, and Cooper D. 2011. Supporting the transition from hospital to home for premature infants using integrated mobile computing and sensor support. *Personal Ubiquitous Comput.*, 15, 8 (December 2011), 871–885. DOI:https://doi.org/10.1007/s00779-011-0402-4
- [7] Gohlke K, and Hornecker E. A Stretch-Flexible Textile Multitouch Sensor for User Input on Inflatable Membrane Structures & Non-Planar Surfaces. *The 31st Annual ACM Symposium on User Interface Software and Technology Adjunct Proceedings (2018)*, pp. 191–193.aa.
- [8] Noh G, Shin J, Cho J, and Bianchi A. 2014. Guidelines for an infant's nutrition monitoring system. In *Proceedings of HCI Korea (HCiK '15)*. Hanbit Media, Inc., Seoul, KOR, 92–97.
- [9] Sareen H, Umapathi U, Shin P, Kakehi Y, Ou J, Ishii H, and Maes P. 2017. Print-flatables: Printing Human-Scale, Functional and Dynamic Inflatable Objects. In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems (CHI '17)*. Association for Computing Machinery, New York, USA, 3669–3680. DOI:https://doi.org/10.1145/3025453.3025898
- [10] Ou J, Skouras M, Vlavianos N, Heibeck F, Cheng C-Y, Peters J, and Ishii H. 2016. AeroMorph – Heat-sealing Inflatable Shape-change Materials for Interaction Design. In *Proceedings of the 29th Annual Symposium on User Interface Software and Technology (UIST '16)*. Association for Computing Machinery, New York, USA, 121–132. DOI:https://doi.org/10.1145/2984511.2984520
- [11] Kim, Seoktae, Kim H, Lee B, *et al*. Inflatable Mouse: Volume-Adjustable Mouse with Air-Pressure-Sensitive Input and Haptic Feedback. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (2008)*, pp. 211–224.
- [12] Croes MJG, Feijs LMG. 2015. Family arizing. In Chen L, Djajadiningrat T, Feijs LMG, Hu J, Kufin SHM, Rampino L, Rodriguez E, and Steffen D, editors, *Aesthetics of Interaction: Dynamic, Multisensory, Wise: Design and Semantics of Form and Movement (DeSForM 2015)*, October 13–17, 2015, Milano, Italy. s.l.: s.n. 2015., pp.

- 318–319
- [13] Hauser S, Suto MJ, Holsti L, Ranger M, and MacLean KE. 2020. Designing and Evaluating Calmer, a Device for Simulating Maternal Skin-to-Skin Holding for Premature Infants. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems (CHI '20)*. Association for Computing Machinery, New York, USA, 1–15. DOI:<https://doi.org/10.1145/3313831.3376539>
 - [14] Röddiger T, Beigl M, Köpke M, and Budde M. 2018. VOCNEA: sleep apnea and hypopnea detection using a novel tiny gas sensor. In *Proceedings of the 2018 ACM International Symposium on Wearable Computers (ISWC '18)*. Association for Computing Machinery, New York, USA, 226–227. DOI:<https://doi.org/10.1145/3267242.3267273>
 - [15] Chen W, Bambang Oetomo S, Tetteroo D, Versteegh F, Mamagkaki T, Serras Pereira M, Janssen L, and van Meurs A. 2015. Mimo pillow – an intelligent cushion designed with maternal heartbeat vibrations for comforting newborn infants. *IEEE Journal of Biomedical and Health Informatics*, 19, 3: 979–985. <https://doi.org/10.1109/JBHI.2014.2349153>
 - [16] Guanyun Wang, Tingyu Cheng, Youngwook Do, Humphrey Yang, Ye Tao, Jianzhe Gu, Byoungkwon An, and Lining Yao. 2018. Printed Paper Actuator: A Low-cost Reversible Actuation and Sensing Method for Shape Changing Interfaces. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems (CHI '18)*. Association for Computing Machinery, New York, NY, USA, Paper 569, 1–12. DOI:<https://doi.org/10.1145/3173574.3174143>
 - [17] Lining Yao, Jifei Ou, Chin-Yi Cheng, Helene Steiner, Wen Wang, Guanyun Wang, and Hiroshi Ishii. 2015. BioLogic: Natto Cells as Nanoactuators for Shape Changing Interfaces. *Proceedings of the 33rd Annual ACM Conference on Human Factors in Computing Systems*. Association for Computing Machinery, New York, NY, USA, 1–10. DOI:<https://doi.org/10.1145/2702123.2702611>
 - [18] Lining Y, Niiyama R, Ou J, Follmer S, Della Silva C, and Ishii H. 2013. PneuUI: pneumatically actuated soft composite materials for shape-changing interfaces. In *Proceedings of the 26th Annual ACM Symposium on User Interface Software and Technology (UIST '13)*. Association for Computing Machinery, New York, USA, 13–22. DOI:<https://doi.org/10.1145/2501988.2502037>
 - [19] Xinlei Z, Shtarbanov A, Zeng J, Chen VK, Bove VM, Maes P, and Rekimoto J. 2019. Bubble: Wearable Assistive Grasping Augmentation Based on Soft Inflatables. In *Extended Abstracts of the 2019 CHI Conference on Human Factors in Computing Systems (CHI EA '19)*. Association for Computing Machinery, New York, USA, Paper LBW2313, 1–6. DOI:<https://doi.org/10.1145/3290607.3312868>